

BSc Project Screening Form: Guidelines

Part 1 – Project Proposal

Student Name	Kushal Kafle
Student Number	2133095
Degree Pathway	Bsc (Hons) Computer Science & Software Engineering
Supervisor Name	Krishna Aryal
Title of Project	Smart Irrigation System
Abstract of the project	This research focuses on uses of IoT and AI in the development of smart systems that can also be implemented in the agricultural development to improve the irrigation system. The devices are to be assembled and integrated with the remote AI models, as per the requirement of this project research. As the traditional irrigation system faces many climatic challenges, and is also not very efficient when it comes to water resource management. This research tends to find an innovative approach to improve and optimize the water management in agriculture with the help of the project based on AI and IoT. The project includes the development of an irrigation system that is scalable and can be adapted to various agricultural settings. One of the goal of this research is finding the right sensors and AI models that can fit the requirements of project. It is also necessary to research on how the water usage analytics can be shown to the end user. This includes finding the right mathematical approach to calculate and estimate the water flow from the system and find out how much water is used by specific crop for the time being. The system interacts with various sensors to collect real time data on soil moisture, temperature and humidity. Then the collected sensor data is analysed by the AI model to make predictions on whether to start or stop the irrigation. Then system measures the total amount of water used for irrigation and produces analytics on the water used by various plants and in various years. Farmers can utilize this analytics and make informed decisions on which plants to grow in the future iterations and estimate the amount of water required for the plant. This ensures that the crops receive the optimal amount of water required for the proper growth, which leads to better crop productions. Keywords: ESP32 based Irrigation; AI in irrigation; Smart Irrigation; Internet of Things in Agriculture; Predictive Irrigation System;

Project deliverables	• Web Application for remote monitoring. • IoT based Demonstration • Final Report • Academic Poster
Description of your artefact	Background: According to IOSR Journal (2015, pp. 32-36), With the water requirements in irrigation being large, there is a need for a smart irrigation system that can save about 80% of the water. Smart Irrigation can also prove to be efficient in Agricultural fields, Lawns & Parks. It is achieved by installing sensors in the field to monitor the soil temperature and soil moisture which transmits the data to the microcontroller for estimation of water demands of plants. According to Choudhary et al. (2019), The autonomous irrigation system used in the study uses real time soil moisture data and AI based prediction algorithms to optimize water usage. The system analyses the historical weather data to predict the rainfall and climate change for more accurate water distribution. Tests conducted in controlled environment showed 80% accuracy which makes is effective solution for water conservation and improves the irrigation efficiency. According to Zhao et al. (2017), Water resource management is crucial in 21st century, as traditional agricultural irrigation systems are inefficient. IoT has introduced smart irrigation as a solution. This paper proposes a LoRa-based smart irrigation system, consisting of a LoRa communication module, solenoid valve, and hydroelectric generator. Data is transmitted wirelessly to the cloud through LoRa gateways, and the system can be controlled remotely via mobile applications. Experimental results demonstrate that the system is reliable in terms of both transmission distance and energy efficiency. According to Pernapati (2018), Since traditional irrigation methods consume a lot of water, there is a need of smart irrigation techniques to minimize the water waste. Smart irrigation system using IoT ans smart wireless sensors is implemented. Environmental conditions around the plants are monitored by the humidity, temperature and moisture sensors, and the ultrasonic sensor measures the water level in the reservoir. The system in controlled by ESP8266 NodeMCU microcontroller which processes the sensor data and then communicates via MQTT protocol to manage water usage efficiently. According to Sitharthan et al. (2023), An autonomous irrigation system is implemented using AI and 6G enabled IoT for smart agriculture. The prediction algorithm is used in the

microprocessor. The system uses historical weather data and IoT to predict rainfall and climate changes and then irrigates the crops accordingly. The system also considers soil moisture level to optimize the water supply. The system was tested with an accuracy of 86.34%, sensitivity of 89.28%, and precision of 91%.

Problem Statement:

This project addresses the challenges faced by existing traditional irrigation systems which are inefficient and do not consider the real time environmental conditions causing over irrigation leading to depletion of water resource or under irrigation leading to decreased crop productions. The project tries to solve the challenges faced by agricultural sectors due to global warming and climate change leading increased concerns about food security and water scarcity.

Aim

To research about the appropriate methods and approaches using IoT and AI in order to minimize the water wastage in agriculture and to increase agricultural productions to tackle global water scarcity and food security.

Objectives

- Research and implement one of the approach or methods for smart irrigation using AI and IoT.
- Build a functional irrigation system using ESP32 to control water flow based on real-time sensor inputs.
- Implement an AI model to analyse the sensor data and predict the irrigation schedule.
- Measure the overall water usage in the given period by specific crops.
- Dashboard for monitoring the system as well as viewing the water usage analytics.

List of features

- Automatic irrigation water flow based on AI predictions.
- Dashboard for remote monitoring of irrigation schedule and water usage analytics.
- Track the plant age in days and irrigate accordingly
- Option for manually entering the crop type and age.

	Added value that the project provides → Supports further research in context of smart irrigation as it provides real world validation and testing. → Research points out the findings about the feasibility and challenges while implementing the smart irrigation in the real world. → The system tracks the water volume in each irrigation and gives detailed water usage analytics. → Remotely monitor the environmental conditions. → The system is cost effective as it uses components that are easily available at cheaper rate. → The system requires only one time installation or setup, and then it runs without any human intervention. Intellectual challenges involved → Sensor Integration and Calibration: Integrating various sensors like soil moisture, temperature and humidity with ESP32 at different parts of the field and calibrating them for accurate data collection, and also handling the noise and errors in sensors in real time ensuring reliability at different environmental settings poses a technical challenge. → Environmental Factors: Since the system operates in open field, predicting the weather and rainfall is challenging due to their unpredictable nature, even weather forecasts are not reliably accurate. So the sudden environmental changes like rainstorms and fluctuating temperature may disrupt the irrigation schedule requiring AI model to quickly adapt and re-estimate the water needs dynamically which is a challenging task.												
Risk Analysis	<table><tr><th>Risk</th><th>Impact</th><th>Likelihood</th><th>Mitigation Strategy</th></tr><tr><td>Risk of Data Quality</td><td>High</td><td>High</td><td>It involves providing data validation checks, implement multiple sensors with options of data redundancy, and periodic sensor calibration.</td></tr><tr><td>Risk of Inaccurate</td><td>High</td><td>High</td><td>This can be mitigated by using multiple resources</td></tr></table>	Risk	Impact	Likelihood	Mitigation Strategy	Risk of Data Quality	High	High	It involves providing data validation checks, implement multiple sensors with options of data redundancy, and periodic sensor calibration.	Risk of Inaccurate	High	High	This can be mitigated by using multiple resources
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	weather forecast Risk of Power Outage Medium High Risk of sensor fault High Medium	for the weather forecast. Also, a hybrid approach with human and AI based prediction may be applied for the weather forecast. Implement backup power sources, use uninterruptible power supplies or solar powers. Implement sensor redundancy by utilizing different sensors for the validation of data, and perform regular maintenance along with calibration of the sensors.
How does your project relate to your degree course and build upon the units/knowledge you have studied/acquired	Throughout my Bachelor degree or course, I acquired a wide range of skill sets and knowledge in various fields. The Smart AI and IoT based irrigation system is built upon several areas of my studies. It integrates the concepts from AI, software engineering methodologies and programming. Principle of Programming My foundation on programming built by this course. It taught java programming in depth, which later helped me to further understand other programming languages and quickly learn them. It also helped me understand how programs communicate with the databases to store and retrieve the data. Fundamental Of Computer Studies In this course I learned how the computer hardwares work at a fundamental level. This course also taught low level programming at basic level which fascinated me even more to dive deep into microcontrollers programming. This is directly applicable to my current project while working with ESP32 microcontrollers and integrating sensors. OOP and Software Engineering	

	This course taught me software development lifecycle, design patterns and various data structures and algorithms. I also gained knowledge about cloud computing. Materials taught in this course are directly relevant to my project Smart Irrigation System. Research Methodologies and Emerging Technologies I learned about the systematic research and development aspect of the project in this course. I gained the knowledge to perform literature review, evaluate the emerging and innovative technologies and apply the appropriate and relevant research methods in the project I am developing. The Smart Irrigation System is built using the latest innovation in sustainable agricultural sector, AI and IoT. In conclusion, my academic courses has built my foundations in AI, computer hardwares, programming, software designing and engineering and software development. Each of these knowledge directly contributes to the development of Smart Irrigation System that is robust, adaptable, scalable and innovative solution for modern and sustainable agriculture.
Resources	Hardware Requirements → ESP32 microcontroller → Soil Moisture Sensor → Temperature and humidity sensor → Water Pump → Relay Module → Remote server with 4GB ram, and Intel core I5 or amd ryzen 5 for AI model Software Requirements → Python IDE → Python Interpreter 3.11 → AI and ML libraries (Scikit-learn, tensorflow, XGBoost, logistic regression, binary classification etc) → Arduino IDE → ESP32 drivers Languages Used → C/CPP for ESP32 programming → Python 3.11 → Postgresql (Database) → ReactJS (for frontend interface)

Have you completed & submitted your ethics form?	☒ YES	☐ NO
If the project is a development of previous work by yourself or others, give details below. Failing to declare such previous work here may be treated as an academic offence		

Supervisor Signature: *Kyala*
29 October 2024

After the proposal has been signed off by both the supervisor and course coordinator scan the proposal and upload on BREO with signatures. Projects that follow proposals that have not been approved may be cancelled and there will be no compensation for any time lost

Part 2 – List of relevant resources

1. Books

- a. Kernighan, B.W. and Ritchie, D.M., 1988. *The C programming language*. 2nd ed. Englewood Cliffs, N.J.: Prentice Hall.
- b. Blum, R., 2014. *Arduino programming in 24 hours, Sams teach yourself*. Indianapolis: Sams Publishing.
- c. Barr, M., 1999. *Programming embedded systems in C and C++*. Sebastopol, CA: O'Reilly Media.

2. Journal Papers

- a. IOSR Journal of Electronics and Communication Engineering (IOSR-JECE) e-ISSN: 2278-2834,p- ISSN: 2278-8735.Volume 10, Issue 3, Ver. II (May - Jun.2015), PP 32-36 www.iosrjournals.org; (Online)
Available
at :<https://smartfasal.in/wp/wp-content/uploads/2019/09/Smart-Irrigation-System.pdf>
[Accessed on: 2024-10-02]
- b. Choudhary, S., Gaurav, V., Singh, A. and Agarwal, S., 2019. Autonomous crop irrigation system using artificial intelligence. *International Journal of Engineering and Advanced Technology (IJEAT)*, 8(5S), pp.46-50. (Online) Available at:
<https://pdfs.semanticscholar.org/f4ec/2797766bd54505f7bbf9d4298d3765846ade.pdf>
[Accessed on: 2024-10-04]
- c. Zhao, W., Lin, S., Han, J., Xu, R. and Hou, L., 2017. LoRa-based smart irrigation system. In: *2017 IEEE Globecom Workshops (GC Wkshps)*, 4-8 Dec. 2017, Singapore. Piscataway: IEEE. (Online)
Available at: <https://ieeexplore.ieee.org/abstract/document/8269115>
[Accessed on: 2024-10-04]
- d. Pernapati, K., 2018. Smart irrigation system using IoT and wireless sensors. In: *2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT)*, 20-21 April 2018, Coimbatore, India. Piscataway: IEEE. (Online) Available at: <https://library-guides.ucl.ac.uk/harvard/journal-article>

[Accessed on: 2024-10-04]

- e. *Sitharthan, R., Rajesh, M., Vimal, S., Saravana Kumar, E., Yuvaraj, S., Abhishek Kumar, Raglend, J.I. and Vengatesan, K., 2023. A novel autonomous irrigation system for smart agriculture using AI and 6G enabled IoT network. Microprocessors and Microsystems, 101, p.104905. (Online) Available at: <https://www.sciencedirect.com/science/article/abs/pii/S0141933123001497>*

[Accessed on: 2024-10-04]

- f. *Agile Alliance (2023) Agile 101: What is Agile?. Available at: <https://www.agilealliance.org/agile101/> (Accessed: 25 October 2024)*